

The normalized trace on a finite CAR stage

MathlibAnnex.CStarAlgebra.CAR.stageTrace

def

The normalized matrix trace provides a positive tracial functional compatible with the CAR inclusions.

Statement

For $n \in \mathbb{N}$, let $A_n = M_{2^n}(\mathbb{C})$ with its operator norm. The finite-stage trace is the continuous complex-linear functional $\tau_n : A_n \rightarrow \mathbb{C}$ whose value is $\tau_n(a) = 2^{-n} \sum_{i=0}^{2^n-1} a_{ii}$. The factor 2^{-n} makes $\tau_n(I) = 1$.

Definition

Take the normalized diagonal sum as a linear map. Each diagonal entry satisfies $|a_{ii}| \leq \|a\|$, so the triangle inequality gives $|2^{-n} \sum_i a_{ii}| \leq 2^{-n} 2^n \|a\| = \|a\|$. This bound supplies continuity. Positivity follows by expanding the diagonal of a^*a into sums of squared absolute values; the trace identity follows by interchanging the two finite summations. Amplification repeats every diagonal entry twice, exactly cancelling the change in normalization.

Assumptions

The index n may be zero, in which case $A_0 = \mathbb{C}$. No representation, choice of vector, or limiting operation enters this definition.

Conclusion

The functional is bounded by $|\tau_n(a)| \leq \|a\|$, is positive on a^*a , and satisfies $\tau_n(ab) = \tau_n(ba)$. Under the standard CAR inclusion $s_n : A_n \rightarrow A_{n+1}$, given by $a \otimes I_2$ and coordinate reindexing, $\tau_{n+1}(s_n(a)) = \tau_n(a)$. These are the separately proved properties cited below.

The root projection E_{00} has trace 2^{-n} ; the root-coordinate functional $a \mapsto a_{00}$ instead takes value 1 there. Thus the trace and the product-vector functional serve different purposes. The declaration here defines a finite-stage functional, not a new assertion about a trace on the completion.

Main citations

- Definition and its exact construction — MathlibAnnex.CStarAlgebra.CAR.stageTrace
- Normalized diagonal sum — MathlibAnnex.CStarAlgebra.CAR.stageTraceLinear
- Entrywise estimate giving continuity — MathlibAnnex.CStarAlgebra.CAR.norm_stageTraceLinear_le
- Bound for the continuous trace — MathlibAnnex.CStarAlgebra.CAR.norm_stageTrace_le
- Normalization at the identity — MathlibAnnex.CStarAlgebra.CAR.stageTrace_one
- Positivity on squares — MathlibAnnex.CStarAlgebra.CAR.stageTrace_star_mul_self_nonneg
- Cyclicity of the finite trace — MathlibAnnex.CStarAlgebra.CAR.stageTrace_mul_comm
- Compatibility with amplification — MathlibAnnex.CStarAlgebra.CAR.stageTrace_step

- Trace of the distinguished projection —
MathlibAnnex.CStarAlgebra.CAR.stageTrace_rootProjection

Lean source signature (exact)

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noncomputable def stageTrace (n : ℕ) : Stage n → L[ℂ] ℂ :=
  (stageTraceLinear n).mkContinuous 1 fun a ↦ by
    simp only [one_mul] using norm_stageTraceLinear_le n a
```

In the source	Mathematical meaning
$n : \mathbb{N}$	The stage index $n \geq 0$, including $n = 0$.
Stage n	The matrix algebra $A_n = M_{2^n}(\mathbb{C})$ with its operator norm.
Stage $n \rightarrow L[\mathbb{C}] \mathbb{C}$	A continuous complex-linear functional $\tau_n : A_n \rightarrow \mathbb{C}$.
stageTraceLinear n	The map $a \mapsto 2^{-n} \sum_{i=0}^{2^n-1} a_{ii}$, so its value at a matrix is its normalized diagonal sum.
$(\dots).mkContinuous 1 \text{ fun } a \mapsto \dots$	Keep that same map and supply $ \tau_n(a) \leq \ a\ $ for every $a \in A_n$; the bound 1 makes the linear functional continuous.
norm_stageTraceLinear_le $n a$	The bound on the normalized sum used in this definition. Positivity, the trace identity and compatibility with later stages are separately cited properties, not additional inputs to this definition.

Further source notes: The formula for its underlying linear map and the proofs of its properties are separate exact citations.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.stageTrace

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